Review of smart systems in aquaculture and aquaponics

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***Abstract*—In this paper, we present a set of works between 2019-2025 in the field of aquaculture and smart aquaculture systems, which witnessed remarkable developments by integrating modern technologies such as artificial intelligence, Internet of Things and cloud computing. The works were based on a set of standards to monitor and automatically control environmental changes in the system and maintain its stability, and the works presented a set of technologies in the field of tracking the growth of crops including fish and plants and enhancing the efficiency of resource management to reduce waste. In addition to works that included proactive prediction of environmental variables, decision support systems for failure analysis, and automated feeder systems, with the aim of developing smart systems that address new challenges and solve previous shortcomings observed in the field.**

Keywords— aquaponics, Aquaculture, Internet of Things, Artificial Intelligenc , Smart Systems

I. Introduction

With the spread of wars and global crises and the spread of food crises and scarcity, it has become important to provide solutions that achieve an economic alternative to deal with these crises. Aquaponics systems are one of the solutions and the most important food production systems at the present time, as they include the economy of resources in a system consisting of two units, an aquaculture system and an irrigation and agricultural fertilization system within a closed system that achieves environmental balance as the wastewater of the aquaculture system is rich in nutrients useful for the plant, but manual labor and reliance on traditional methods are expensive and not protected from human errors, and therefore this field is witnessing a major shift in the methods used and with the rapid technological development, the integration of artificial intelligence and Internet of Things(IoT) technologies has become a necessity to achieve. In this context, this research reviews the latest innovations between 2019-2025 that provided modern technologies that support the field of aquaculture and aquaculture in order to build an intelligent system capable of dealing with challenges and achieving positive results.

II. Comparative analysis of previous works

A. Monitoring and control using the Internet of Things

Internet of Things (IoT) technologies are among the most important monitoring technologies in smart systems today in the fields of aquaculture, irrigation, and aquaponics systems. Several modern technologies can be observed in Table 1 in the field of smart monitoring, including sensors, cloud platform support, and microcontrollers such as Arduino and NodeMCU. The aim is to identify and address problems as quickly as possible and reduce human intervention. In addition, integrating the work of [14]the WeMos D1 Wi-Fi Uno (ESP8266) microcontroller for data processing and transmission via Wi-Fi. The MQTT protocol was employed in the study by [13]with the aim of ensuring secure communication between devices and the cloud to enable the integration of artificial intelligence and to automatically improve system standards, while platforms like Ubidots and Thingspeak were used in the studies of [11] and [12] for data collection and real-time display. Additionally, the use of solar energy in [13] was beneficial for the environment and energy economy, especially in farms. The works also included, such as the work of [10], the construction of alert systems that send alerts via SMS to notify any system malfunction. The results of these techniques presented in Table 2 showed the potential to develop smart monitoring and control systems that contribute to real-time fault handling.

TABLE. 1 PREVIOUS BUSINESS TECHNOLOGIES AND TOOLS FOR REMOTE MONITORING AND CONTROL IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Algorithms and Processing Techniques	Control Devices
Bakar et al., 2022[10]	SMS alert system (GSM)	Arduino Mega
Haryanto et al., 2019[11]	Data visualization using the Ubidots platform	NodeMCU
Ntulo et al., 2021[12]	Data collection and analysis via Thingspeak	Arduino Uno
Khaoula et al., 2021[13]	MQTT protocol for data analysis and security	NodeMCU with solar power systems
Abdullah et al., 2022[14]	Remote control using the Blynk application	WeMos D1 Wi-Fi Uno (ESP8266)

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TABLE. 2 THE CONTRIBUTION OF BUSINESSES TO REMOTE MONITORING AND CONTROL IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Key Results
Bakar et al., 2022 [10]	Stable pH range (6.5–8) with consistent plant growth improvement.
Haryanto et al., 2019 [11]	High sensor accuracy, enhanced lettuce and fish growth despite heat challenges.
Ntulo et al., 2021[12]	Strong relationship between time and plant height and fish weight ($R^2 > 0.96$).
Khaoula et al., 2021 [13]	System integration using solar power and improved sustainability.
Abdullah et al., 2022 [14]	Reduced labor dependence with enhanced plant and fish productivity.

In the presented monitoring and control systems, sensors are important elements. In previous works in Table 6, it was observed that most studies used temperature and humidity sensors and pH sensors to ensure an ideal environment for plants and fish, and as a key tool for monitoring water quality. Water level sensors were also used to avoid problems of water shortage or excess. Works such as [13] distinguished themselves by using CO_2 and ammonia sensors to monitor the quality of nutrients in the water.

TABLE. 3 COMPARISON OF SENSORS USED IN MONITORING AND CONTROL FOR PREVIOUS WORKS IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Temperature & Humidity Sensor	pH Sensor	Water Level Sensor	Soil Moisture Sensor	Sensors Ammonia	Sensors CO_2
Bakar et al., 2022 [10]	✓	✓	✓	✗	✗	✗
Haryanto et al., 2019 [11]	✓	✓	✓	✗	✗	✗
Ntulo et al., 2021 [12]	✓	✓	✓	✗	✗	✗
Khaoula et al., 2021 [13]	✓	✓	✓	✗	✓	✓
Abdullah et al., 2022 [14]	✓	✗	✓	✓	✗	✗

Figure 3 illustrates the architecture of the Internet of Things (IoT) system used for monitoring in the aquaponics systems of [11] work. It consists of three layers: first, the physical layer includes sensors that capture data; then, the gateway layer consists of a NodeMCU device as an intermediary for transmitting data via the MQTT (Message Queuing Telemetry Transport) protocol. The data is transmitted via MQTT/WebSocket and directed to the MQTT Broker, then to the IoT Cloud for storage and analysis. Then comes the application layer, which includes a user interface that allows data to be displayed in various forms.

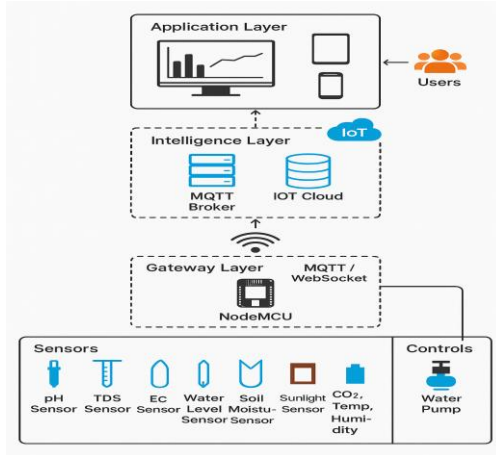

Fig. 1.The surveillance system and the Internet of Things in operation [11].

B. *Automated and intelligent control of Aquaponics systems*

To enhance the efficiency of aquaculture and aquaponics systems, many studies have proposed various initiatives that include the technologies outlined in Table 3. With the aim of building an intelligent system that operates with automated control, strategic work has been developed to improve water quality and maintain the environment of organisms using techniques that mostly included fuzzy logic due to its ability to adapt and handle imprecise and vague data in the environment and traditional techniques that rely on experts and precise equations, such as PID (Proportional-Integral-Derivative) control, contributed to achieving good results in maintaining the environment of the aquaponics system. Additionally, the work of [1] and [2] included a biofilter for the growth of specialized nitrifying bacteria in the nitrification process, which is the conversion of ammonia (NH_3) or ammonium (NH_4) to nitrate (NO_3-) to improve the nutrients in the water. The most important task of the system was to provide a medium for the life of these bacteria and to maintain the quality of the nutrient solution to contribute to the health of the plant. [8] work presented ideas to improve water conditions using Genetic Algorithms (GA) to predict the best water conditions, as well as Random Forest modeling to respond to plant growth based on water criteria. The operations enhanced their systems by integrating with the Internet of Things and cloud platforms to process and analyze data in the best ways. MATLAB Simulink was used for simulation and applying mathematical algorithms, AWS Cloud for remote monitoring, Raspberry Pi as the main processing unit for local analysis, and Arduino ESP8266 for wireless communications. On the other hand. The results in Table 4 show that these intelligent systems have significantly improved the system's performance. These systems have also contributed to the effective control of resources, productivity, and the health of organisms.

TABLE. 4 SMART CONTROL TECHNIQUES AND TOOLS IN PREVIOUS WORKS IN AQUACULTURE AND AQUAPONICS SYSTEMS

Title and Authors	Algorithms	Processing Technology	Control Mechanism
Bracino et al., 2020[1]	Fuzzy Logic	MATLAB Simulink	Electric valves and cooling/heating elements
Kok et al., 2024[2]	PID Control Algorithm	Cloud-based data analysis	Control of pumps and other environmental elements
Concepcion et al., 2020[3]	Fuzzy Logic using Mamdani	Fuzzification and defuzzification using Centroid	Electric valve control
Pitowarno et al., 2023[4]	Fuzzy Logic	Data analysis via IoT network	Control of pumps, valves, and heaters
Nagothu et al., 2025[5]	Fuzzy Logic	Cloud-based data transfer (AWS)	Control of actuators (pumps, aerators)
Tsai et al., 2022[6]	Fuzzy Control System	Arduino and Raspberry Pi	Air pumps and automatic feeding system
Johan & Suharjito, 2023[7]	Fuzzy Logic (Decision-making), IoT	Arduino ESP8266, Raspberry Pi	Actuators for temperature, aeration, and water flow
Ula et al., 2024[8]	Random Forest , Genetic Algorithm	IoT with integrated sensors	Automatic feeding and control systems
Artetxe et al., 2024[9]	Fuzzy Logic for irrigation control	LoRa-enabled microcontroller	Water pump for precision irrigation

TABLE. 5 BUSINESS CONTRIBUTION TO SMART MONITORING AND CONTROL IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Results
Bracino et al., 2020 [1]	Stabilized pH (7-8) and temperature (25-30°C) to optimize the environment for nitrifying bacteria.
Kok et al., 2024 [2]	Water quality stabilization, resource efficiency improvement, and error reduction using PID.
Concepcion et al., 2020 [3]	Maintained optimal pH (6.0-6.5) and EC (0.8-1.2 mS/cm); enhanced plant growth.
Pitowarno et al., 2023 [4]	Restored water quality (temperature: 32°C, pH: 6-8) efficiently and quickly; improved fish productivity.
Nagothu et al., 2025 [5]	Achieved 98% accuracy; remote monitoring via mobile app; enhanced aquatic organisms' environment. And it demonstrated the superiority of fuzzy logic over ANFIS and ANN algorithms.
Tsai et al., 2022 [6]	Increased shrimp survival rate by 33.3% compared to traditional methods.
Johan & Suharjito, 2023 [7]	Increased shrimp survival by 6%, improved growth rate by 28%.
Ula et al., 2024[8]	Stable water conditions, enhanced productivity.
Artetxe et al., 2024 [9]	Reduced water usage by 23.1%, with an additional 4.07% savings by integrating weather forecasts.

In smart systems, sensors are used to monitor and control key environmental factors. We presented a comparison of works on sensors in Table 5. It can be seen that all presented studies included pH and often temperature sensors, indicating their importance in the study of water quality. The works that introduced aquaponic systems also include EC sensors to monitor nutrient levels, as in [3]. Advanced systems, such as those developed by [5], included sensors for dissolved oxygen (DO), turbidity, and salinity for a good assessment of the environment.

TABLE. 6 COMPARISON OF SENSORS USED IN SMART CONTROL OF PREVIOUS WORKS IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	pH Sensor	Temperature Sensor	Turbidity Sensor	EC Sensor	TDS Sensor	Water Level Sensor	DO Sensor	Salinity Sensor	Soil Moisture Sensor
Bracino et al., 2020 [1]	✓	✓	✗	✗	✗	✗	✗	✗	✗
Kok et al., 2024 [2]	✓	✓	✓	✗	✓	✓	✗	✗	✗
Concepcion et al., 2020 [3]	✓	✗	✗	✓	✗	✗	✗	✗	✗
Pitowarno et al., 2023 [4]	✓	✓	✓	✗	✗	✗	✗	✗	✗
Nagothu et al., 2025 [5]	✓	✓	✓	✗	✗	✗	✓	✓	✗
Tsai et al., 2022 [6]	✓	✓	✗	✗	✗	✓	✗	✗	✗
Johan & Suharjito, 2023[7]	✓	✓	✗	✗	✗	✗	✓	✗	✗
Ula et al., 2024[8]	✓	✓	✗	✗	✗	✗	✗	✓	✗
Artetxe et al., 2024 [9]	✓	✗	✗	✗	✗	✗	✗		✓

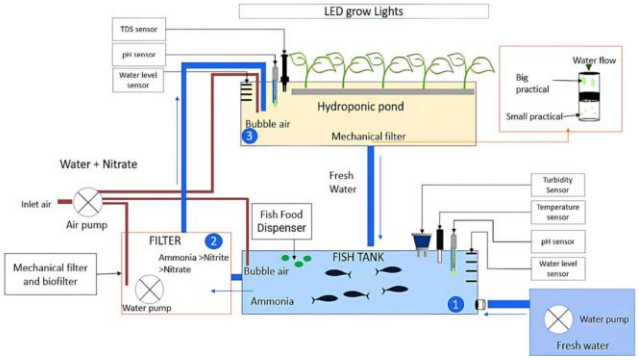

Fig. 2.Aquaponics model for work [2]

In Figure 2, the aquaponics system from [2] Fish waste is converted by a mechanical and biological filter into nitrates that nourish the plants, while the plants purify the water to be recycled for the fish. And water and air pumps to ensure continuous flow and aeration, in addition to a food dispenser for feeding the fish, and sensors used for monitoring. The system is intelligently controlled through a PID algorithm, which adjusts the system when there is a deviation from the ideal values. LED lighting enhances plant growth.

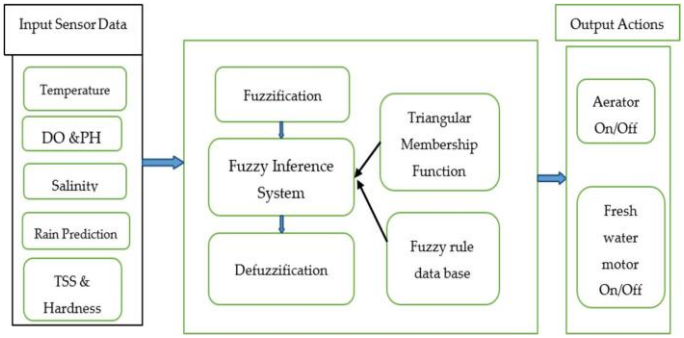

Fig. 3.The mechanism of automatic control using fuzzy logic in the operation of [5].

Figure 3 illustrates the decision-making model by [5], based on fuzzy logic used in some of the mentioned studies. The dataset captured by the sensors undergoes a fuzzification phase by converting the data into fuzzy values, followed by fuzzy inference based on rules and fuzzy functions, and then defuzzification to convert the results into clear decisions. The outputs include turning on/off the aerator or the fresh water pump.

C. Decision support in aquaculture and aquaponics systems

Decision support and optimization techniques are an important axis in intelligent systems as they use a set of technologies that include data analysis and transformation into models that help in making informed decisions. Table 7 represents part of these technologies in the field of aquaponics and aquaculture. For example, [15] focused on developing a system using artificial intelligence and the Internet of Things to support positive decision-making in aquaculture farms by collecting data on water quality and nutrition levels. A deep learning model (DNN) was developed using Bayesian optimization to tune parameters, predict fish growth, and control feeding In contrast, [17] presented an intelligent cultivation system based on analyzing farmers' needs using QFD, which facilitated farm management and achieved user satisfaction [16] developed a decision support system based on digital twins to simulate aquaponic systems in real-time. The system aims to coordinate distributed urban agriculture systems to improve production and reduce resource waste. Algorithms such as linear regression, support vector machines, and decision trees were applied to predict the growth rates of plants and fish. The work relied on the use of the sensors represented in Table 9 for monitoring and data collection. The contribution of the businesses can be observed in Table 8. Figure 4 shows the design of [16] aquaponics system, which includes a set of elements that intervene in making smart decisions. Among these are the inputs, represented by the sensors used for monitoring, and the outputs, which include control devices for regulating pH levels and solutions to increase or decrease acidity, as well as an air pump. The design also features biological components such as a biological filter and nitrification bacteria to convert ammonia to nitrite (Ammonia > Nitrite > Nitrate) for irrigating plants.

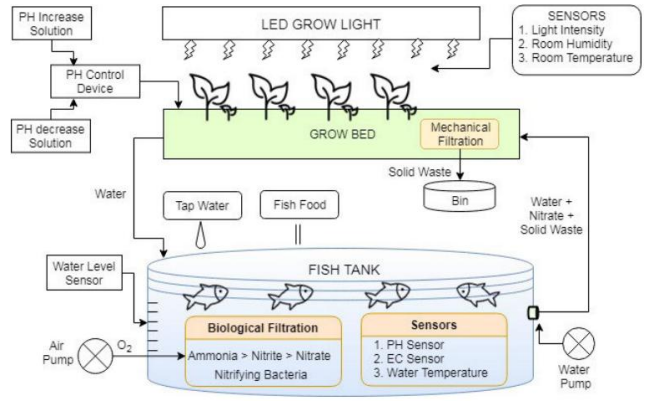

Fig. 1. Aquaponics model for work [16]

TABLE. 7 DECISION SUPPORT TECHNIQUES AND TOOLS IN AQUACULTURE AND AQUAPONICS SYSTEMS FOR PREVIOUS WORKS

Study	Algorithms	Processing Units	Control
Chiu et al., 2022[15]	Deep Neural Networks (DNN) with Bayesian Optimization	Arduino Mega2560 with Wi-Fi module	Water pump, heater, agitator, smart feeder
Ghandar et al., 2021[16]	Digital Twin Simulation	IoT and cloud computing platforms	Water pumps, lighting system (LEDs), flow control valves
Setiawan & Surantha, 2024[17]	QFD (Quality Function Deployment), IoT	IoT Sensors, Cloud Computing	Automatic feeders, real-time monitoring systems

TABLE. 8 THE CONTRIBUTION OF BUSINESSES TO SUPPORTING DECISION-MAKING IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Results
Chiu et al., 2022 [15]	Achieved high prediction accuracy for fish growth (R^2 = 0.94), reduced feed waste, improved water quality monitoring.
Ghandar et al., 2021 [16]	Enhanced urban aquaponics optimization, reduced resource waste, and real-time performance predictions.
Setiawan & Surantha, 2024 [17]	Improved water quality, increased production efficiency, user satisfaction above 4/5.

TABLE. 9 COMPARISON OF THE CRITERIA USED IN DECISION-MAKING FOR PREVIOUS PROJECTS IN AQUACULTURE AND AQUAPONICS SYSTEMS.

Study	Dissolved Oxygen Sensor (DO)	Temperature Sensor	Turbidity Sensor	pH Sensor	Proximity Switch	Soil Moisture Sensor	Light Sensor (LDR Shield)	Water Flow Sensor	Total Dissolved Solids (TDS) Sensor	Salinity
Chiu et al., 2022 [15]	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗
Ghandar et al., 2021 [16]	✗	✓	✗	✓	✗	✗	✓	✓	✓	✗
Setiawan & Surantha, 2024 [17]	✗	✓	✗	✓	✗	✗	✗	✗	✗	✓

D. Techniques for improving aquaponics systems and smart aquaculture

Improving aquaponics and hydroponics systems using modern technologies aims to maintain system quality, enhance productivity, and conserve resources. Some works have presented solutions and techniques that can be observed in Table 10 along with the standards in Table 0. [18] presented the DRAPS system, which achieved exceptional efficiency in resource consumption and high productivity through a dual water recirculation system that separates fish tanks and plant cultivation, supported by artificial intelligence technologies. On the other hand, [19] provided solutions for sustainability by integrating a Recirculating Aquaculture System (RAS) and a Zero Water Discharge (ZWD) system, relying on the measurement of Temperature, Dissolved Oxygen, and Ammonia, which resulted in reduced water discharge and energy consumption while maintaining high survival rates for shrimp. On the other hand, [20] Improving nutrient concentration using a machine learning algorithm to reduce harvest time and enhance lettuce growth. Table 11 contains the results of this work, as they reflect good solutions in resource efficiency and production improvement, with each focusing on distinctive aspects of smart and sustainable aquaponics systems.

TABLE. 10 TECHNIQUES AND TOOLS FOR IMPROVING AND DEVELOPING AQUACULTURE AND AQUAPONICS SYSTEMS FOR PREVIOUS PROJECTS

Title	Algorithms	Processing Units	Control Units (e.g., pumps)
Karimanzira & Rauschenbach, 2021[18]	LSTM (Anomaly detection), Bayesian (Fault identification)	DRAPS	Dual-circulation pumps
Kassem et al., 2021[19]	N/A	Hybrid RAS-ZWD System	Intelligent aeration and feeding systems
Dhal et al., 2022[20]	Linear SVM (Classification), PCA (Feature selection),	Nutrient Data Analysis	Balance improvement recommendations via control

TABLE. 11 THE CONTRIBUTION OF BUSINESSES TOWARDS IMPROVING AND DEVELOPING AQUACULTURE AND AQUAPONICS SYSTEMS.

Title	Results
Karimanzira & Rauschenbach, 2021 [18]	20% productivity improvement, 97% reduction in water consumption.
Kassem et al., 2021 [19]	27% energy saving, 90% shrimp survival rate.
Dhal et al., 2022 [20]	Increased crop size and quality, reduced harvest time to 4 weeks.

E. Predictive and intelligent analysis models

Prediction systems are used to analyze environmental data using techniques such as machine learning and deep learning. In the field of ecoponics, several works have made various contributions, such as the work of [21], which included classifying the growth stages of plants like lettuce by reducing the number of features and testing deep algorithms. In addition to the work of [23], which relied on criteria such as pH and temperature. To predict the types of fish capable of living in a specific environment. There are works that have focused on predicting water quality, such as [22], which aimed to predict water quality and detect "harmful algal blooms." To prevent its spread and thus improve salmon farms. And also the work of Rasheed [24] ,which supported the idea of predicting water quality parameters in real-time by reducing computation time. The algorithms and data used can be observed in Tables 12 and 13.

TABLE. 12 COMPARISON OF FORECASTING TECHNIQUES IN PREVIOUS WORKS ON AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Dataset Size	Data
Lauguico et al., 2020[21]	300 images	Morphological features
Eze et al., 2023[22]	10 weeks	Chlorophyll-a (470 nm, 530 nm), turbidity
Islam et al., 2021[23]	191 samples	pH (6.5–9), temperature, turbidity
Rasheed Abdul Haq & Harigovindan, 2022[24]	Multiple datasets (ADAK, MAC)	Salinity, pH, dissolved oxygen (DO), temperature

TABLE. 13 COMPARISON OF THE DATA USED IN FORECASTING PREVIOUS WORKS

Study	Algorithms	Performance Results (Accuracy)	Key Metrics
Lauguico et al., 2020 [21]	LR, KNN, L-SVM	KNN: 91.67%	Accuracy, Cross-validation
Eze et al., 2023 [22]	EEMD-MLR-LSTM	MAE: 0.0375, MAPE: 0.0072	MAE, RMSE, MAPE
Islam et al., 2021 [23]	Random Forest, J48, KNN	RF: 88.48%	Accuracy, Kappa
Rasheed Abdul Haq & Harigovindan, 2022 [24]	CNN-LSTM, CNN-GRU	CNN-LSTM outperformed all models	MAE, MSE, RMSE

F. Intelligent and automated feeding systems

Intelligent feeding systems are a key element of hydroponics and a vital axis for organism survival. They also significantly reduce resource economy and human error. [25]) created a smart feeding system that employs load cell sensors to measure exact weight and the ARIMA model to forecast future feeding requirements in order to increase feeding accuracy in aquaponics systems. In [26] introduced an automated feeding system based on fuzzy logic, taking into account environmental factors such as temperature and pH, using ultrasound to monitor water and feeding schedules. [27] study integrated computer vision with YOLOv5 and. fuzzy neural networks (FNN) to effectively analyze the feeding behavior of fish in aquaculture systems. The techniques used can be referenced in Tables 14.

TABLE. 14 COMPARISON BETWEEN SMART SYSTEMS FOR FEEDING AQUATIC ORGANISMS

Study	Objective	Algorithms	Performance Results
Risma et al. (2025)[25]	Improve feeding accuracy using ARIMA and IoT	ARIMA	±0.34g feeding accuracy, 123.12% growth rate
Surbakti et al., 2025[26]	Optimize feeding using fuzzy logic	Fuzzy Logic	Improved fish health, reduced mortality
Zhou et al., 2023[27]	Integrate computer vision with FNN	YOLOv5 + FNN	80-90% accuracy in feeding behavior analysis

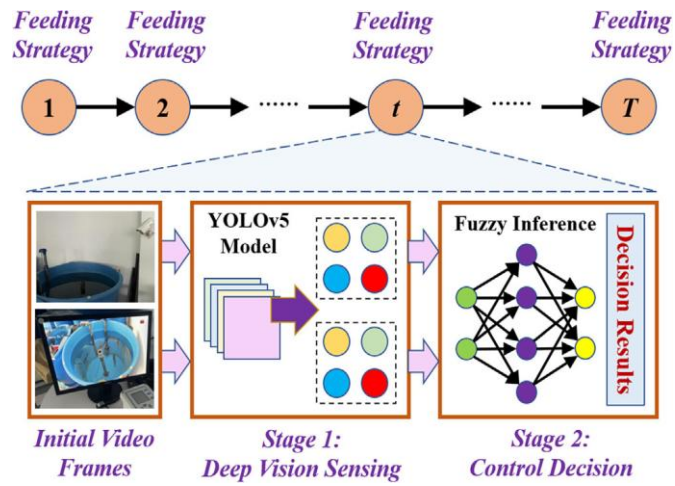

Fig. 2. Intelligent feeding control framework using deep learning and fuzzy logic for [27] work

Figure 5 illustrates the decision-making stages in a system [27]. In the first stage , deep vision sensing is performed. The inputs are characterized as videos, which are divided into frames. The YOLOv5 model is then used to analyze each frame to identify objects and important activities. The second stage involves a fuzzy inference system using YOLOv5 results and fuzzy rules to output automated control decisions in the system.

III. DISCUSSION

Previous research on hydroponics and aquaculture systems offers advanced solutions using smart and advanced technologies to improve production efficiency, resource consumption, and reduce human errors in several areas, including in the field of intelligent control, which work in this area shows the importance of maintaining environmental variables in a stable degree, such as improving water quality and ensuring nutrient quality, with the support of technology and cloud analytics to enable real-time monitoring and minimize human intervention. Decision support systems using models such as digital twins, neural networks, and utilizing a previous data set to achieve a high ability to identify positive decisions by analyzing data and predicting environmental conditions to ensure the efficiency and safety of the system. Smart feeding systems ensure the health of organisms and reduce the risk of overfeeding. Techniques such as computer vision and predictive models are used to determine feeding times and minimize the waste of resources. The new optimization technique focused on several areas, including managing resource consumption and improving productivity such as plant growth rate and crop quality. Predictive modeling in intensive system management also played an important role in predicting water quality and growth and fighting harmful algae to ensure stable environmental conditions. In Table 0 you can see the sensors used in each area of smart systems in aquaculture and aquaponics.

Sensor	Smart Control	IoT Monitoring & Control	Decision Support	Smart Feeding	Techniques for improving	Predictive
pH Sensor	✓	✓	✓	✓	✓	✓
Temperature Sensor	✓	✓	✓	✓	✓	✓
Turbidity Sensor	✓	✓	✓	✗	✗	✓
EC Sensor	✓	✓	✗	✗	✗	✗
TDS Sensor	✗	✓	✗	✗	✗	✗
DO Sensor	✓	✓	✓	✗	✓	✓
Salinity Sensor	✗	✓	✓	✗	✓	✗
Humidity Sensor	✗	✓	✗	✗	✗	✗
CO_2 Sensor	✗	✓	✗	✗	✓	✗
Ammonia Sensor	✗	✓	✗	✗	✓	✗
Water Flow Sensor	✗	✗	✓	✗	✗	✗
Light Sensor	✗	✗	✓	✗	✗	✗
Load Cell	✗	✗	✗	✓	✗	✗
RTC	✗	✓	✗	✓	✗	✓
Ultrasonic Sensor	✗	✗	✗	✓	✗	✗
Vision Sensors (YOLOv5)	✗	✗	✓	✓	✗	✓

IV. Conclusion

Aquaponics and smart aquaculture systems stand out as strategic solutions to achieve food security and raise the quality of production. These systems provided several solutions that included achieving environmental stability through monitoring systems, as well as automatic control systems that contributed to reducing human errors and adapting to environmental variables, and these systems also supported methods in improving the system, including improving nutrients, controlling resources and monitoring crop growth, in addition to systems that support farmers' decisions capable of adapting to their needs with failure analysis within the system. Also, solutions have been presented about nutrition in aquaculture systems that were aimed at saving resources while supporting these ideas with environmental forecasting techniques that enable them to anticipate risks proactively even with these ideas, but there is a missing element that is observed that these works are missing some important data regarding the study of nutrients, not considering the importance of studying ammonia, nitrates and solids as they are important in aquaponics systems related to the life of aquatic organisms. These substances may cause toxicity in the water of the aquaponics system, which leads to system failure, and the systems that focus on nutrient recovery are few, which affects the sustainability of the system and its need for other fertilizers, and therefore this topic is still a subject of study if there is a need to create alternative solutions that are more efficient, more comprehensive in environmental standards, more accurate results, and adaptability to achieve the new economic revolution.

V. References

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